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POSTER

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MAXIM SIDOROV, Ulm University, Ulm, Baden-Wurttemberg, Germany

WOLFGANG MINKER, Ulm University, Ulm, Baden-Wurttemberg, Germany

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Emotion Recognition and Depression Diagnosis by Acoustic and Visual Features: A Multimodal Approach

Maxim Sidorov
Institute of Communications Engineering
University of Ulm
Ulm, Germany
maxim.sidorov@uni-ulm.de

Wolfgang Minker
Institute of Communications Engineering
University of Ulm
Ulm, Germany
wolfgang.minker@uni-ulm.de

ABSTRACT

There is an enormous number of potential applications of the system which is capable to recognize human emotions. Such opportunity can be useful in various applications, e.g., improvement of Spoken Dialogue Systems (SDSs) or monitoring agents in call-centers. Depression is another aspect of human beings which is closely related to emotions. The system, that can automatically diagnose patient's depression can be helpful to physicians in order to support their decisions and avoid critical mistakes. Therefore, the Affect and Depression Recognition Sub-Challenges (ASC and DSC correspondingly) of the second combined open Audio/Visual Emotion and Depression recognition Challenge (AVEC 2014) is focused on estimating emotions and depression. This study presents the results of multimodal affect and depression recognition based on four different segmentation methods, using support vector regression. Furthermore, a speaker identification procedure has been introduced in order to build the speaker-specific emotion/depression recognition systems.

Categories and Subject Descriptors

H.4 [**Information Systems Applications**]: Miscellaneous; D.2.8 [**Software Engineering**]: Metrics—*complexity measures, performance measures*

Keywords

Emotion and depression recognition, audio-visual features, feature-based fusion, support vector regression

1. INTRODUCTION

Automatic recognition of human emotions and even depression based on audio and visual signals is in the focus of research groups all over the world. While enabling machines to recognize human affects may be useful in various applications, the recognition performance is still not satisfying. Meanwhile, computer systems are already deeply involved in recognizing, interpretation, or even synthesis of human emotions. For instance, Affective Computing [13] is one of the subject which deals with human affects.

Despite the fact that emotions have subjective nature, humans have a fine-tuned emotional model of people they know allowing for recognizing their emotional states more accurate. Furthermore, the basic emotions are shared between all people and cultures [16]. There is even a baseline psychology- and physiology-based continuous measurements forming *Russell's Emotion Circumplex* [14] which cover the wide range of human emotions.

Whereas the majority of research concentrates on seven emotional primitives (neutral, anger, fear, joy, sadness, boredom, and disgust) while recognising human emotions, we focused on estimation of continuous-space measurements. The following measurements, related to human emotions, are under examination in the study: *Arousal*, *Dominance*, and *Valence*. Arousal is the activity level of central nervous system, showing the rate of individual's preparedness to act in current situation. Valence is intrinsic attractiveness or suffering of an event, situation, or object. Dominance is the individual's sense of how much they feel to be in control of the current situation [11, 27]. It should be noted that the mentioned emotional primitives can be easily derived from the two of three analysed in this study continuous dimensions (Arousal and Valence). Thus, in our investigations we deal with regression machine-learning tasks.

There is an increasing number of scientific papers, where the authors consider different ways of emotion recognition. Since the authors may use different corpora or performance metrics, these approaches are not always comparable. Therefore, such challenges as AVEC provide a general framework by using that researches from all over the world can objectively compare their achievements and set up state-of-the-art approaches. It was done during the first [20], the second [21], and the third [28] Audio/Visual Emotion recognition Challenges (AVEC), the first and the second Emotion Recognition in The Wild Challenges (EmotiW) [7], the first international audio/visual mapping personality traits (MAPTRAITS), and Computational Paralinguistics Challenge (ComParE) [19].

Each approach on multimodal affect recognition addresses four key components: the audio and the video feature extraction procedure, the modelling algorithm and the fusion techniques. Scientific groups use the wide variety of approaches and their combinations.

For the audio part, all variety of Low-Level Descriptors (LLDs) and their functionals have been used for emotion

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recognition, based on their successful applications on such problems as automatic speech recognition, speaker and gender identification. LLDs of speech signals can be extracted using Praat software [4]. Several functionals-based feature sets have been specially developed for emotion recognition challenges. Thus, feature sets of 384 and 1,582 functionals have been proposed at the INTERSPEECH 2009 Emotion challenge [17] and at the INTERSPEECH 2010 Paralinguistic Challenge [18] correspondingly. The latest audio-based feature set consisting of 6,373 features has been suggested for the ComParE [19]. All of these features can be easily extracted out of speech waveforms using openSMILE software [9]. The authors in [15] used three different audio feature sets for affect recognition: 16 MFCCs with their delta and acceleration (48), energy and spectral features (34), and 2268 functionals which have been derived from LLDs. In [5] the authors have achieved reasonable results in emotion recognition using the following set of audio-based features: the means, the standard deviations, the ranges, the maximum values, the minimum values and medians of the pitch and the energy. Regarding the learning stage of emotion recognition using audio cues, the most common algorithms used are Multi Layer Perceptron (MLP) [25] and Support Vector Machine (SVM) [12], as well as their combination [30].

Classical feature sets for the visual part including Local Binary Patterns (LBP) [1], Gabor, LPQ, PHOG [6] and their combination [26] have been successfully utilised for facial expression recognition and face detection. Recently suggested LPQ-TOP and LGBP-TOP[2] have also been successfully used for emotion recognition. Concerning the learning phase of visual-based emotion, affect and depression recognition the most common method used is SVM for both classification and regression tasks.

Several fusion techniques may be utilised for merging the multimodal information. Thus, audio-visual affect and depression recognition are based on three types of fusion methods: feature-level (also called early fusion), decision-level and model-level [22, 31]. *Feature-level* fusion can be implemented by combining the features of different modalities into a single feature set. Feature-level fusion has the disadvantage of including the different kinds of modalities, which have different metric levels. Furthermore, high dimensionality of resulting feature set can become the other problem. In *decision-level* fusion, the multimodal data are treated independently and the single-modal hypothesises are combined at decision level using some criterion (weighted sum, etc.). Decision-level fusion overcomes the problems of the feature-level fusion but it is based on the assumption that the data for different modalities are independent, that is not necessary true in reality. The Hidden Markov Model, in which different components are responsible for different data sources, can be an example of *model-level* fusion approach.

We present the multimodal emotion and depression recognition system based on SVM trained by Sequential Minimal Optimization (SMO) algorithm [10], and feature-level fusion technique. Four segmentation methods have been explored and corresponding results have been demonstrated for the both tasks.

The rest of the paper is organized as follows: Section 2 describes AVEC'14 corpus; Section 3 reviews the audio and video feature extraction methods as well as segmentation techniques; the proposed system is described in Section 4; Section 5 presents the experiments and results obtained with both the development and the test sets; finally, Section 6 presents the conclusions and future work.

2. CORPUS DESCRIPTION

A subset of AVEC'13 consisting of 150 videos is used for the challenge. Only two tasks in a human-computer interaction scenario have been selected to include in AVEC'14 dataset. During *Northwind* scheme participants read aloud an extract of the story "The North Wind and the Sun" in German. *Freeform* task is participant's answers to several general questions such as "What was your best gift, and why?", again in German [27].

In order to obtain the level of depression, participants have been asked to fill in a standard self-assessed depression questionnaire (the Beck Depression Inventory-II, BDI-II, [3]) consisting of 21 questions. The BDI-II scores are from 0 to 63 and can be interpreted as different ranges of depression: from minimal to severe depression. It should be noted that only one depression label has been set for each recording.

Each affect dimension (Arousal, Dominance, and Valence) has been annotated separately by minimum of three and maximum of five raters. The process of annotation has been performed using a two-axis joystick so that every recording's second has 30 continuous values. Final labels for every recording are average values over all annotators.

Highlighting the dependence between depression and emotions the authors emphasise a non-linear correlation between the affect and depression labels.

3. FEATURES

The recordings in the challenge are quite long, therefore segmentation technique should be applied. The organisers of the challenge have been realized multimodal audio-visual features for variety of segments in order to perform experiments. A brief description of the suggested features and corresponding segmentation methods are in the following Section.

3.1 Audio Features

The audio part of the recordings has been characterised by 2,268 dimensional feature vector. The following set of audio features has been selected to represent the challenge recordings: energy- and spectral-based LLDs (32 x 42 functionals), voicing LLDs (6 x 32 functionals), delta of energy/spectral features (32 x 19 functionals), delta of voicing features (6 x 19 functionals), and voiced/unvoiced duration features (10). The LLDs cover the state-of-the-art features of emotion recognition such as: loudness, MFCCs (1-16), jitter, shimmer, energy in different bands, probability of voicing, etc. Functionals features comprise the most descriptive functional dependences of LLDs: standard deviation, quartiles, linear regression slope and quadratic regression coefficient a with corresponding approximation errors, mean, maximum, minimum, etc.

Four segmentation schemes have been proposed in order to conduct the experiments. *Per_instance* segmentation uses only one average feature vector per recording. The second segmentation, called *vad_seg*, uses a voice activity detector [8] in order to analyse speech activity. Then, the audio features have been calculated on corresponding segments where speech activity has been detected. Pauses of more than 200 ms have been used to split speech activity sections. The

Table 1: Results for affect recognition. Development partition. Performance is measured in Pearson's correlation coefficient.

Task	Features	Audio			Video			Audio-Video		
		Val.	Arous.	Dom.	Val.	Arous.	Dom.	Val.	Arous.	Dom.
Northwind	per_instance	0.4336	0.5991	0.5281	0.2651	0.1893	0.4038	0.6153	0.6673	0.6222
	vad_seg	0.2050	0.4283	0.4655	0.3685	0.4942	0.4944	0.3925	0.4352	0.4883
	long	0.3572	0.7184	0.5298	0.3317	0.2167	0.1728	0.6336	0.6526	0.6343
	short	0.1328	0.3664	0.2655	0.4745	0.2806	0.1592	0.5751	0.4551	0.5086
Freeform	per_instance	0.3231	0.2391	0.2530	0.4229	0.1398	0.0209	0.5424	0.4803	0.5150
	vad_seg	-0.0656	0.0590	-0.1265	0.2696	0.1196	0.2627	0.1858	0.3309	0.1661
	long	0.1445	0.3590	0.1798	0.1550	0.2898	0.2041	0.4348	0.7018	0.3073
	short	0.0412	0.2032	0.0791	0.1906	0.1051	0.1351	0.3750	0.4399	0.2253

third and the fourth segmentation methods consider overlapping segments of 3 and 20 seconds shifted towards at a rate of one second (*short* and *long* schemes, correspondingly).

3.2 Video Features

For the visual part, the local dynamic appearance descriptor LGBP-TOP from the eMax face analysis toolbox [2] has been utilised as video-based features. The original video recordings have been resampled to uniform 30 frames per second at 640 x 480 pixels. Further, face localisation and segmentation have been performed with the publicly available Viola & Jones [29] face detector. The following parameters set of LGBP-TOP feature extraction procedure has been used: 18 filters with variable orientation and frequencies, but constant amplitude and phase; the fixed window of 5 overlapping frames. It should be noted that only features extracted from XY image planes have been included into feature set.

4. EMOTION AND DEPRESSION PREDICTION WITH MULTIMODAL FEATURES

The Support Vector Machine (SVM) learned by the Sequential Minimal Optimization (SMO) algorithm [23] has been deployed for solving the affect and the depression regression tasks.

For the performance evaluation in case of affect recognition the sample Pearson's correlation coefficient COR has been used. Let y_k and $\hat{y}_k$ be the ground-truth and predicted values, correspondingly, COR can be defined as:

$$COR = \frac{\sum_{k=1}^{N}(y_k - \mu_{yk})(\hat{y}_k - \mu_{\hat{y}k})}{\sqrt{\sum_{k=1}^{N}(y_k - \mu_{yk})^2}\sqrt{\sum_{k=1}^{N}(\hat{y}_k - \mu_{\hat{y}k})^2}} \quad (1)$$

where N is the number of predictions, μ_{yk} and $\mu_{\hat{y}k}$ are the sample mean of ground-truth and predicted values, respectively.

In case of depression recognition, the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE) have been chosen as the main performance criteria:

$$RMSE = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(y_k - \hat{y}_k)} \quad (2)$$

$$MAE = \frac{1}{N}\sum_{k=1}^{N}(y_k - \hat{y}_k) \quad (3)$$

There are ground-truth labels for Arousal, Dominance and Valence measurements every 1/30 s; as the audio-based functional features have been obtained at different rates, average values of corresponding labels have been calculated in order to equal the rates of the labels and the feature vectors. The same procedure has been performed for the video-based features. Since there is only one ground-truth label per recording for the depression sub-challenge, the same depression label has been duplicated so that the number of features and the labels has become the same.

Firstly, the baseline speaker-independent affect and depression recognition have been performed. Secondly, in order to extend the advantage of using the speaker-dependent classification-based emotion recognition [24, 25] to regression-based affect and depression recognition, a speaker identification approach has also been applied. The following subsections describe the process of SVM application for considering tasks more deeply.

4.1 Speaker-Independent Affect and Depression Recognition

To perform the baseline speaker-independent experiments the four proposed segmentation methods have been evaluated. To fit the labels into the features for the affect recognition sub-challenge, the labels have been averaged correspondingly. Thus, in case of *per_instance* segmentation method, only one average label has been set as a target for the regression task. For the *short* segmentation scheme, every 90 labels, with the constant overlapping, have been averaged and set as a label. Analogically, the similar procedure has been applied for *vad_seg* and *long* methods.

For the depression recognition sub-challenge the given label has been multiplied for all segments of the corresponding recording.

Firstly, audio and video features have been treated separately, then the feature-based fusion technique without further feature selection or reduction methods has been utilised.

In order to rescale the attributes of different units and scales, the statistical normalization procedure (also called Z-transformation) has been applied as a preprocessing step.

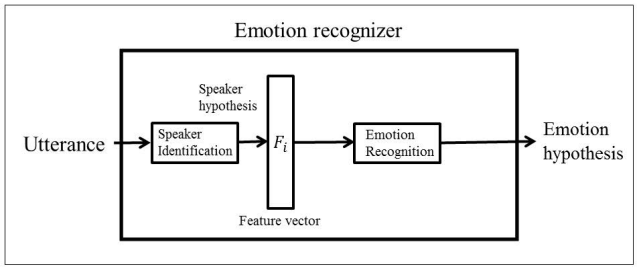
To produce the affect and the depression hypothesises, the SVM instance has been trained with the training portion and

Table 2: Results for depression recognition. Development partition. Performance is measured in mean absolute error (MAE) and root mean square error (RMSE) over all sequences.

Task	Features	Audio		Video		Audio-Video	
		MAE	RMSE	MAE	RMSE	MAE	RMSE
Northwind	per_instance	9.2712	11.5184	24.1634	31.7911	7.6383	**9.6481**
	vad_seg	9.7814	12.1308	11.1988	**13.8724**	9.7096	11.4782
	long	8.6122	10.3761	26.3424	34.0621	7.9479	10.2044
	short	8.6942	**10.3255**	17.5607	21.7821	8.8702	10.8245
Freeform	per_instance	7.9044	9.9045	21.7123	28.1306	7.7487	9.4623
	vad_seg	11.3031	15.3112	14.8427	**17.6672**	11.8038	13.6414
	long	7.4193	**9.7495**	48.0671	66.7377	7.2447	**8.9636**
	short	9.5687	12.9427	17.9564	23.4683	9.4319	12.6752

evaluated with the development set. Hence, one single SVM has been created addressing the emotions of every speaker in the database.

Figure 1: Speaker-dependent affect recognition system.



4.2 Speaker-Dependent Affect and Depression Recognition

Incorporating speaker-specific information into the emotion/depression recognition process may be done in many ways. A very straight forward way is to add this information to the set of features. The approach results in a two-stage procedure (see Figure 1): First, the speaker is identified and then this information is included into feature set directly. Both emotion/depression recognition and speaker identification hybrid systems have been investigated and evaluated in this study.

To examine the theoretical improvement of using speaker-specific information for affect and depression recognition, the true information about the speaker has been used. Then a real speaker identification component has been applied. For both tasks (affect/depression recognition and speaker identification), SVM trained by SMO algorithm has been chosen as a modelling method.

In the first experiment, the focus was on investigating the theoretical improvement, which may be achieved using speaker-based adaptivity. For this, known speaker information (true labels) has been used. In the system, the speaker information was simply added to the feature vector. Hence, all utterances with the corresponding speaker information have been used to create and evaluate the SVM-based emotion and depression models.

Additionally, a second experiment was conducted including a real speaker identification module instead of using known speaker information. First, an SVM-based speaker identifier was created during training phase. Furthermore, the known speaker information was included into the feature vector for the training of the emotion/depression model. The testing phase starts with the speaker identification procedure. Then, the speaker hypothesis was included into the feature set which was in turn fed into the emotion/depression recognizer. In contrast to the first experiment, these experiments are not free of speaker identification errors.

In order to generate more statistical significant results, the complete regression process has been performed using 6-fold cross-validation procedure with the *speaker-based stratified sampling* for each task and experiment.

5. EXPERIMENTAL RESULTS

The results of speaker-independent affect and depression recognition investigations are in Table 1 and in Table 2, correspondingly.

Regarding the affect recognition, it is obvious that using multimodal approach results in much higher performance values than the single-modality approaches. Interesting that the performance of the Arousal and Dominance recognition was higher using the audio-based features, whereas the video-based Valence recognition outperformed the audio-based approach. Nevertheless, it should be noted that the results based on different segmentation methods are not comparable, since the performance values have been calculated on different levels.

Concerning the depression recognition, all the same experiments have been conducted, but in this case the evaluation metrics were MAE and RMSE (see Table 2). Again, the highest performance, in terms of RMSE, has been achieved by using the multimodal approach. Since all segments of the same recording have the same labels too, we applied an aggregation technique in order to produce only one depression hypothesis for each recording. Thus, only on average hypothesis has been calculated over all segment-based assumptions. The final hypothesis of each recording has been set as the nearest to the average hypothesis whole number. The aggregation technique allows to produce comparable performance values, since all the criteria have been calculated on the same level. An average value (over four segmentation methods, and two tasks) of RMSE for the multimodal approach is 10.8 against 11.5 and 29.7 in case of audio- and

Table 3: Results for speaker-dependent affect recognition. Train plus development sets, 6-fold cross-validation with speaker-based stratified sampling. Performance is measured in Pearson's correlation coefficient. Average values over all labels.

Task	Features	Audio			Video			Audio-Video		
		baseline	real SP	SVM SP	baseline	real SP	SVM SP	baseline	real SP	SVM SP
Northwind	vad_seg	0.4470	**0.5228**	0.4635	0.6741	**0.6832**	0.6704	0.6302	**0.6738**	0.6440
	per_instance	0.5140	0.5174	**0.5212**	0.3786	**0.4536**	0.3974	0.5890	0.5891	**0.5892**
Freeform	vad_seg	0.2133	**0.5753**	0.4057	0.8929	**0.8971**	0.8936	0.7189	**0.7300**	0.7128
	per_instance	0.2602	0.2637	**0.2731**	0.2488	0.3318	**0.3318**	0.5497	**0.5499**	0.5495

Table 4: Results for speaker-dependent depression recognition. Train plus development sets, 6-fold cross-validation with speaker-based stratified sampling. Performance is measured in root mean squared error.

Task	Features	Audio			Video			Audio-Video		
		baseline	real SP	SVM SP	baseline	real SP	SVM SP	baseline	real SP	SVM SP
Northwind	vad_seg	11.2394	**10.2584**	10.4930	5.3799	**4.7081**	4.8216	8.0297	**6.8236**	7.5048
	per_instance	10.2865	10.2522	**10.2061**	27.0421	**15.4807**	15.6965	8.8514	**8.8376**	8.8379
Freeform	vad_seg	15.7375	**8.9371**	11.4724	2.0023	**1.5389**	1.9089	4.4614	5.4771	**3.1699**
	per_instance	10.8020	10.7466	**10.4222**	30.7593	15.7498	**15.4214**	9.1328	9.0369	**8.7005**

video-based modalities, correspondingly. Since the lowest value of RMSE has been achieved using multimodal approach on the *freeform* task, these settings of *per_instance* segmentation have been chosen in order to build a system for the testing partition. The both training and development sets have been used in order to train the SVM-based regression model. The obtained results, namely 10.826 and 8.327 for RMSE and MAE, respectively, slightly outperformed available baseline results (10.859 and 8.857) which have been published by the organisers.

An addition of speaker-specific information into affect and depression recognition procedures increases the performance of the SVM application (see Table 3 and Table 4). There, *baseline* columns correspond to speaker-independent affect and depression recognition procedures, *real SP* columns show the results of using the ground-truth speaker-specific information, and *SVM SP* columns introduce the results of real SVM-based speaker identification procedure incorporating into affect/depression recognition systems. It is clear, that due to speaker identification errors the performance of regression tasks using the real speaker identification component is lower than with ground-truth speaker information. Another disadvantage of the approach is that being a supervised learning procedure, the speaker identification component requires already known speakers. Meanwhile, in case of *per_instance* segmentation method some of the speakers have only a few recordings. This fact prevents those speakers from being into the train and the test sets simultaneously. Nonetheless, the proposed approach allows to find those speakers, whose voices sound similar (or analogous appearance) and to take an advantage of these similarities.

It should be noted that the speaker-based stratified sampling and the same depression labels for each segment of the same recording result in significant lower error values while X-Validation-based depression recognition experiments have been conducted. Moreover, in case of speaker-dependent depression recognition the aggregation technique has not been applied.

6. CONCLUSION

We have presented an audiovisual system for recognition both affect and depression. The modalities have been treated separately, as well as the merged features have been investigated. Four proposed segmentation techniques have also been examined. Further, the speaker identification has been incorporated into the recognition systems. This procedure is able to catch the similarities between speakers and results in performance improvement.

It is evident that already a very simple method as extending the feature vector with additional speaker-specific information can improve the affect and depression recognition performance. This improvement is more significant when using true speaker-specific information. These results are very encouraging leading to further more sophisticated approaches on speaker-dependent emotion and depression recognition, e.g., applying speaker adaptation methods already known from speaker-adaptive speech recognition.

However, for building accurate individual emotion models, balanced databases are required. In order to build emotion model for each speaker, a high number of emotional samples are needed for each speaker.

While SVM already provides reasonable results for speaker identification, we are still examining its general appropriateness. The usage of other-possibly more accurate-identifiers may improve the performance of this hybrid system. In addition, by applying feature selection techniques (principal component analysis, genetic algorithm-based feature selection, etc.) for these problems, further performance improvement could be achieved.

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